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A study of sub-pixel interpolation algorithm in digital speckle correlation method

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ABSTRACT

Sub-pixel interpolation technique is a way that getting sub-pixel gray value by interpolation algorithm and then it is substituted into the correlation expression to calculate the sub-pixel position. It is a key way to improve the displacement measurement accuracy in Digital Speckle Correlation Method (DSCM), meanwhile, it is more time consuming than other steps. A new quarter sub-pixel registration method is proposed which can significantly cut down the sub-pixel searching time. On the premise of not reducing accuracy, an improved Taylor's series expansion in gradient algorithm is used to get higher registration speed. At the same time, a new cubic convolution interpolation method is proposed to get extremely high accuracy. These conclusions are proved by the results of the experimental system.

Keywords: Displacement measurement, Digital speckle correlation method (DSCM), Sub-pixel registration, Interpolation algorithm, Cubic convolution algorithm, Gradient algorithm

1. INTRODUCTION

Digital Speckle Correlation Method (DSCM) was proposed by W. H. Peter and W. F. Ranson[1] from South Carolina University of USA and I . Yamaguchi[2] of Japan independently and synchronously in 1980's. The main idea is to capture the digital speckle images of the measurement object before and after deformation or movement, find the corresponding speckle area in the sample picture (before deformation or movement) and in the target picture (after deformation or movement) by correlation algorithm. So it changes the measurement of the object deformation or movement to the correlation calculation of two digital speckle pictures. This method is not only suit for micro-displacement measurement but also suit for long distance and topography measurement. For the reason of simple data capture, high degree of automation, high accuracy and sensibility and low requirement of measuring environment , this method is developing very fast and widely used in different fields[3] [4]. Sub-pixel registration is the most important way to improve the measurement accuracy of DSCM and is the key technique of DSCM, meanwhile, it is the most time consuming step in DSCM too. There have been many researches in this field from domestic and abroad and many methods been proposed, such as curved surface fitting, interpolation algorithm, frequency correlation, etc. Although it is said that the accuracy of some methods is up to 0.01 pixels in theory, it is not easy to realize the measurement index in practice. This paper is mainly about the interpolation algorithm of sub-pixel registration. A new quarter sub-pixel registration method especially for interpolation algorithm is proposed which can significantly cut down the sub-pixel searching time. After the study of gradient algorithm, an improved gradient algorithm with second order Taylor's series expansion is used to get higher searching speed while not deduce the accuracy. Also a new cubic convolution interpolation method is proposed to get extremely high accuracy. The experimental system is built and according to the analysis of the test results, the comparison study of the previous 3 interpolation methods as well as bilinear interpolation method is made. These conclusions can be used as the selecting guidance of the different interpolation methods in DSCM.

2. SUB-PIXEL INTERPOLATION ALGORITHM

Interpolation method is an effective and high accuracy way to get the sub-pixel gray value of the digital speckle picture.

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Based on these gray values, the two corresponding speckle areas of the measurement object before and after movement can be found out though correlation algorithm. The following is the principle of 4 interpolation algorithms.

2.1 **Bilinear interpolation algorithm**[5]

Suppose that gray value is gradually changing in the image, and the gray value of a point (x, y) can be expressed by the function $F = f(x, y)$. A relatively smooth function can be used to express a smooth or approximate smooth surface. Bilinear interpolation algorithm uses four adjacent pixels to form a continuous surface.

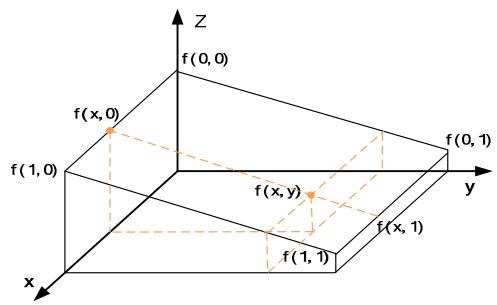

Fig.1 Bilinear interpolation method

The principle of bilinear interpolation algorithm is shown in Fig.1. $f_{(0,0)}$, $f_{(0,1)}$, $f_{(1,0)}$, $f_{(1,1)}$ are gray value of 4 adjacent points in graphic plane. To get the gray value of a sub-pixel point (x, y) among these 4 points, first figure out $f_{(x,0)}$ from $f_{(0,0)}$ and $f_{(1,0)}$, and $f_{(x,1)}$ from $f_{(0,1)}$ and $f_{(1,1)}$, then figure out $f_{(x,y)}$ from $f_{(x,0)}$ and $f_{(x,1)}$. So $f_{(x,y)}$, the gray value of a sub-pixel point (x, y) interpolated in the image can be calculated as follows [6]:

$$\begin{aligned} f_{(x,y)} &= (1-y)f_{(x,0)} + yf_{(x,1)} \\ &= (1-y)[(1-x)f_{(0,0)} + xf_{(1,0)}] + y[(1-x)f_{(0,1)} + xf_{(1,1)}] \\ &= f_{(0,0)} + x(f_{(1,0)} - f_{(0,0)}) + y(f_{(0,1)} - f_{(0,0)}) + xy(f_{(0,0)} + f_{(1,0)} - f_{(0,1)} + f_{(1,1)}) \end{aligned} \tag{1}$$

2.2 **Gradient interpolation algorithm**

Gradient method was used in light flow calculation and was brought into image registration by Freeman [7] firstly. Zhou, etc. [8] applied this method in experimental mechanics to measuring thermal deformation of the electronic package devices. The basic idea of this algorithm is as follows:

Let $f(x, y)$ denote the gray value of point (x, y) in the image. First order Taylor series is used to get the gray value of sub pixel $(x + \Delta x, y + \Delta y)$, ignore the high order parts,

$$f(x + \Delta x, y + \Delta y) = f(x, y) + \Delta x \cdot f_x(x, y) + \Delta y \cdot f_y(x, y) \tag{2}$$

It can be seen from above that how to express the gradient operator is the key to calculate the gray value, Horn operator, Barron operator and Spline operator are commonly used.

Horn operator [7] :

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$$\begin{cases} \dfrac{\partial f(x,y)}{\partial x} = \dfrac{1}{2}\left[f(x+1,y) - f(x,y) + f(x+1,y+1) - f(x,y+1)\right] \\ \dfrac{\partial f(x,y)}{\partial y} = \dfrac{1}{2}\left[f(x,y+1) - f(x,y) + f(x+1,y+1) - f(x+1,y)\right] \end{cases} \tag{3}$$

Barron operator [9] :

$$\begin{cases} \dfrac{\partial f(x,y)}{\partial x} = \dfrac{1}{12} f(x-2,y) - \dfrac{8}{12} f(x-1,y) + \dfrac{8}{12} f(x+1,y) - \dfrac{1}{12} f(x+2,y) \\ \dfrac{\partial f(x,y)}{\partial y} = \dfrac{1}{12} f(x,y-2) - \dfrac{8}{12} f(x,y-1) + \dfrac{8}{12} f(x,y+1) - \dfrac{1}{12} f(x,y+2) \end{cases} \tag{4}$$

Spline operator [10] :

$$\begin{cases} \dfrac{\partial f(x,y)}{\partial x} = \dfrac{1}{8} f(x-2,y) - \dfrac{6}{8} f(x-1,y) + \dfrac{6}{8} f(x+1,y) - \dfrac{1}{8} f(x+2,y) \\ \dfrac{\partial f(x,y)}{\partial y} = \dfrac{1}{8} f(x,y-2) - \dfrac{6}{8} f(x,y-1) + \dfrac{6}{8} f(x,y+1) - \dfrac{1}{8} f(x,y+2) \end{cases} \tag{5}$$

Horn operator is used in this paper because it has simple expression and high calculation efficiency.

Gradient interpolation algorithm takes into account of the gradient changing of the gray value in image, it's easy to implement, but the method has its shortcomings too. Considering the long gradient expansion of binary first-order Taylor expansion

$$f(x+\Delta x, y+\Delta y) = f(x,y) + \Delta x \cdot f_x(x,y) + \Delta y \cdot f_y(x,y) + R_1 \tag{6}$$

R_1 is omitted in the algorithm:

$$R_1 = \frac{1}{2}(\Delta x \frac{\partial}{\partial x} + \Delta y \frac{\partial}{\partial y})^2 f(x+\theta_1 \Delta x, y+\theta_2 \Delta y) \tag{7}$$

In the expression, $0 < \theta_1, \theta_2 < 1$, Δx and Δy are the location of the point that expanded. Clearly, the introduction error will change along with the location of the target point in image, addition, the error will also increase when $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ increase, that is, the higher the frequency of the image, the greater the gradient error.

2.3 Improved gradient interpolation algorithm

Using first-order Taylor series expansion in the interpolation formula can change the complex arithmetic calculation to the simple linear calculation. But if the images are rich of high-frequency information, the error of this method will become bigger. Then the size of the correlation window has to be increased in order to reduce the interpolation error, this will increase the searching time simultaneously. The use of second-order Taylor series expansion in place of the first-order in the expansion of gradient interpolation algorithm can retain more high-frequency information, improve the degree of approximation and effectively reduce the size of the correlation window, so speed up the whole procedure of sub-pixel registration. Using Horn operator as the gradient operator, second-order Taylor series expansion of the binary function is

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$$f(x_0+\Delta x, y_0+\Delta y) \approx f(x_0, y_0) + \left(\Delta x \frac{\partial}{\partial x} + \Delta y \frac{\partial}{\partial y}\right) f(x_0, y_0) + \frac{1}{2!}\left(\Delta x \frac{\partial}{\partial x} + \Delta y \frac{\partial}{\partial y}\right)^2 f(x_0, y_0) \tag{8}$$

2.4 Cubic convolution interpolation algorithm [11][12]

Cubic convolution interpolation algorithm is more commonly used in grid data interpolation, using the nearest 16 pixels to estimate a sub-pixel gray value based on the cubic convolution formula. There are many cubic convolution formulas, some have the effect of low pass filter, and some have the effect of high pass filter. A good formula should achieve a balance between high frequency and low frequency information. At the same time, cubic convolution interpolation has a wide frequency band, is more applicable to the image with wide spectrum.

According to the interpolation theory, cubic convolution expression for equal interval sample of two-dimension image is as following

$$s(x,y) = \sum_l \sum_m c(x_l, y_m) h(|x - x_l|) h(|y - y_m|) \tag{9}$$

In the expression h is interpolation core ; $s(x,y)$ is the sub-pixel point to be interpolated ; $c(x_l, y_m)$ is the sample value at (x_l, y_m). It is clear that the interpolation core is the key of cubic convolution interpolation algorithm.

Cubic convolution interpolation algorithm uses the nearest 16 pixels to estimate the sub-pixel gray value. The following are expressions of several typical interpolation cores in cubic convolution interpolation algorithm:

1) univariate interpolation core [13] :

$$h(x) = \begin{cases} (A+2)x^3 - (A+3)x^2 + 1 & 0 \le x \le 1 \\ Ax^3 - 5Ax^2 + 8Ax - 4A & 1 < x \le 2 \\ 0 & 2 < x \end{cases} \tag{10}$$

In the expression, A is a parameter.

2) bivariate interpolation core:

$$h(x) = \begin{cases} \frac{1}{6}(-9b-6c+12)x^3 - \frac{1}{6}(12b+6c-18)x + \frac{1}{6}(-2b+6) & 0 \le x < 1 \\ \frac{1}{6}(-b-6c)x^3 - \frac{1}{6}(6b+30c)x^2 + \frac{1}{6}(-12b-48c)x + \frac{1}{6}(8b+24c) & 1 \le x < 2 \\ 0 & 2 \le x \end{cases} \tag{11}$$

b, c are variants and (b, c) = (0.33, 0.33) or (1.5, -0.25).

The selection of the parameters of interpolation core [14] :

In specific applications, generally select a single variable interpolation core, this is based of the bull summation idea of Coons-Gordon surface, that is a two-variable interpolation can be fulfilled by appropriate stacking of the single variable interpolations.

The nature of the interpolation core can be assessed in the frequency domain. The ideal interpolation core has unity-gain in band-pass, zero gain in the band-stop area, which can effectively inhibit or pass through the different frequency components of signal.

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For single variable interpolation core A is the only variable parameter, in order to simplify the process of deriving the value of A, assume that the $f(k\Delta t)$ is the sample of function $f(t)$, $k=0,\pm1,\pm2,\cdots$; Δt is the constant sampling frequency, the cubic convolution interpolation can be expressed as:

$$s(t)=\sum f(k\Delta t)\cdot h(t-k\Delta t) \tag{12}$$

Through the analysis of H(ω), the spectrum of h(x), in order to show H(ω) a low-pass filter characteristics, and be as flat as possible in pass band, A should be $-\frac{1}{2}$ or $-\frac{3}{4}$.In condition A=$-\frac{3}{4}$ a certain degree of overshoot can be seen both in pass band and stop band. Therefore, general in practical application A=$-\frac{1}{2}$.

3. QUARTER SUB-PIXEL REGISTRATION METHOD

Appling gray interpolation method in the procedure of sub-pixel registration can improve searching accuracy while reduce the searching speed simultaneously. This method becomes the most time-consuming step in DSCM because of its complicated formula. To overcome this problem a new quarter sub-pixel registration method is proposed which can increase the speed of sub-pixel registration significantly.

Assuming the correlated integer-pixel location figured out by the calculation of the integer pixel searching as (x, y), then the maximum correlated sub-pixel position must be located in a certain rectangle area of (x-1, y+1), (x -1, y - 1), (x+1, y -1), (x +1, y +1). As the area is very small, the correlation coefficient must have the character of 'single peak' in this area. Based on the analysis above, a quarter sub-pixel registration method has been used to locate the maximum correlated position. The searching time can be reduced effectively.

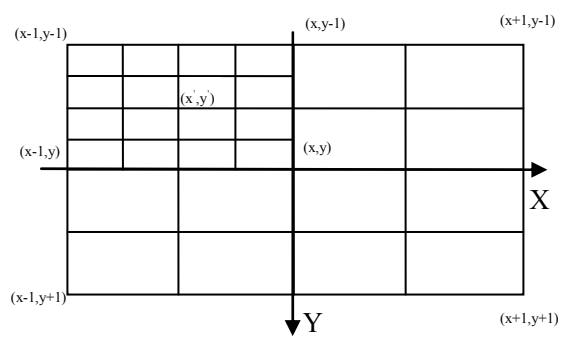

Fig.2 Principle of quarter sub-pixel registration method

The principle of quarter sub-pixel registration method is: equally divide the rectangle area into 4 parts both in direction x and y, by the way of a sub-pixel interpolation algorithm, the gray value of all 25 points can be get, and the correlation coefficient of these 25 points can be get too by the correlation algorithm. Then a new point with the maximum correlation coefficient is chosen as the new center. 4 points around this center forms a new rectangle area. Repeat the steps above to divide and calculate, until the area is small enough and get the final maximum correlation point. In practice, as shown in Fig. 2, the searching way is in direction X first, then direction Y. Firstly fix y, get 5 correlation value in the direction X, find x' with the maximum correlation value, then fix $x=x'$, in Y direction get 5 points, find y' with the maximum correlation value. The rectangle region around (x',y') formed by (x-1, y), (x -1, y -1), (x, y -1) and (x, y) is treated as a new region to be divided and searched. The specific steps as follows (assume the precision is a):

(1) Make sure the initial registration region: left = x-1; top = y-1; right = x +1; bottom = y +1; in which (left, top) and (right, bottom) are the coordinates of the bottom right corner and the top left corner of the registration area;

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(2) Fix $y = \frac{top + bottom}{2}$ first, calculate correlation value of 5 points in X direction to find the maximum value of location $x^{'}$, then fixed $x = x^{'}$, continue searching the maximum correlation value of $y^{'}$ in Y direction;

(3) Set a new registration region around the point $(x^{'}, y^{'})$ formed by (x-1, y), (x -1, y -1), (x, y -1) and (x, y), then change the value of left, top, right and bottom;

(4) When (right-left)> a, or (bottom-top)> a, return to step (2), otherwise, $x_c = \frac{left + right}{2}$, $y_c = \frac{top + bottom}{2}$ is the final sub-pixel location needed.

4. EXPERIMENTAL SYSTEM AND RESULTS

4.1 Experimental system

As shown in Fig.3, the experimental system consists of Speckle object, light source, image acquisition system, displacement calibration equipment and data processing software. Speckle object can move along a guide rail. Image acquisition system consists of CCD, image acquisition card and a computer. CCD is used to capture the digital speckle picture before and after movement (shown in Fig.4); Double frequency laser interferometer is used as displacement calibration equipment. The speckle object moves 0.2mm each time, then the movement of two pictures should be 1.837 pixels by calibration. The movement of speckle object has been consecutive measured for 9 times in experiment, using an area of 100×100 pixels to calculate the correlation value.

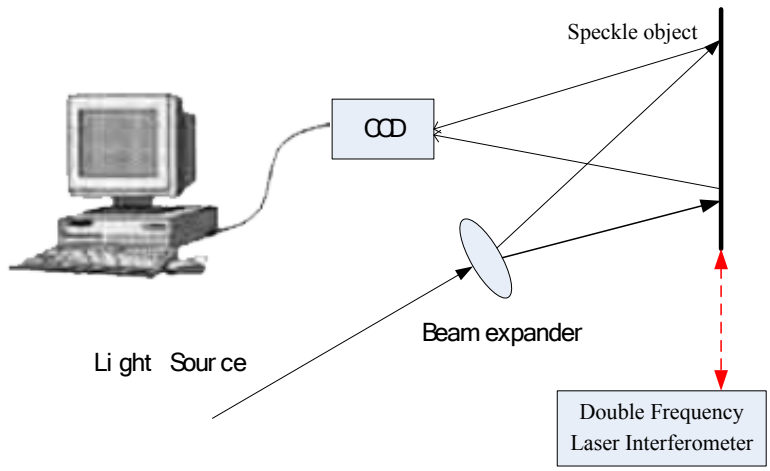
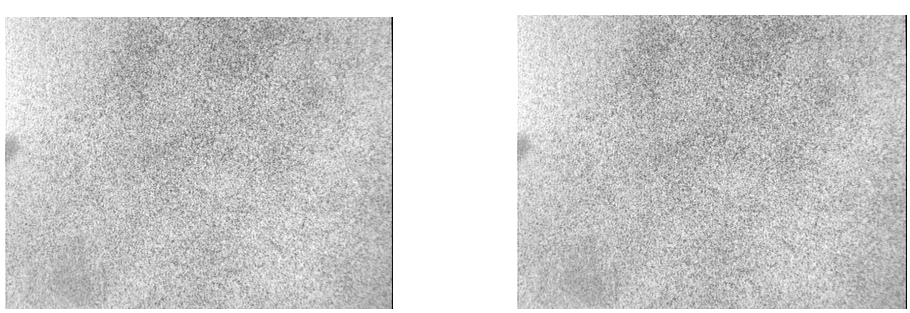


Fig.3 DSCM experimental system



(a) (b)

Fig.4

(a) Digital speckle picture before movement; (b) Digital speckle picture after movement of 0.2 mm

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4.2 The results of the experiment

The results of the experiment are shown as Table 1 and the registration time is shown in Fig.5:

Table.1 Sub-pixel searching result of 4 different methods (Unit is pixel)

	Bilinear	Cubic Convolution	Gradient	Improved Gradient
1	1.864	1.837	1.830	1.829
2	1.863	1.836	1.827	1.826
3	1.864	1.835	1.825	1.824
4	1.862	1.850	1.827	1.825
5	1.877	1.837	1.844	1.842
6	1.863	1.838	1.830	1.829
7	1.866	1.836	1.829	1.828
8	1.863	1.832	1.829	1.828
9	1.860	1.834	1.824	1.824
3σ	±0.029	±0.0052	±0.01	±0.01

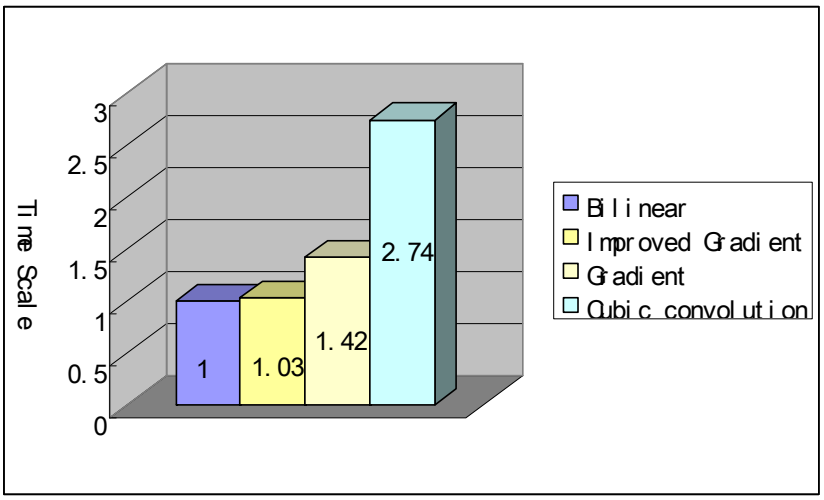

Fig.5 Registration time of 4 different methods

Through the analysis of experimental data of 4 interpolation algorithm above, it can be concluded that bilinear interpolation method has simple formula, highest speed of calculation (in Fig.5, generally assume that the time consumption of bilinear method is 1 in order to be convenient for comparison. The number on the column is the relative time ratio to bilinear method, for example, 1.42 on the gradient column means the registration time of thist method is 1.42 times of that of bilinear method.), but lowest accuracy, the 3σ error is 0.029 pixels. The formula of cubic convolution interpolation method is the most complicate cause it's using the gray values of 16 pixels around the

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interpolation point and therefore needs more time to carry out the sub-pixel registration, but its accuracy is also better than all the other methods, the error is only 0.0052 pixels, searching time is about 2.74 times of bilinear interpolation method on average. For gradient method the error is 0.01 pixels and searching time is 1.42 times of bilinear interpolation method. For the improved gradient interpolation method, the expansion has higher order, so the correlated window can be reduced to 50 × 50 while keep the same precision of 0.01 pixels, so the sub-pixel searching speed can be as quick as bilinear interpolation method (only about 1.03 times of the bilinear interpolation method).

4.3 Error Analysis

In experiments, major sources of error come from CCD noise, environment vibration, and the error of double frequency laser interferometer. The choice of correlation coefficient expression affects the accuracy of methods to some extent. That is, accuracy of the measurement method is limited by the method itself, the quality of the hardware, and environmental noise at the same time.

In order to improve the accuracy and reliability of the measurement, the environmental disturbance need to be controlled and the suitable digital speckle image need to be chosen and experimental equipment with stable performance need to be used.

5. CONCLUSION

As a whole-field, non-contact measurement method, DSCM possesses broad application foreground. Using high resolution CCD, it can get even higher accuracy. In this paper, four sub-pixel interpolation methods, including bilinear interpolation, gradient interpolation, improved gradient interpolation and cubic convolution interpolation algorithms are studied. Efficiency and accuracy of them are compared and analyzed based on the results of the experimental system. Registration time of the bilinear interpolation is the shortest of all, while the error is the biggest. The formula of the new cubic convolution algorithm is the most complicated so that the calculation time is the longest, but the accuracy is the highest and stable. The gradient interpolation is commonly used. The accuracy of this method is higher than the bilinear interpolation while the consuming time is longer. An improved gradient algorithm is proposed in this paper which replaces the first order Taylor's series expansion with the second order. Though the formula is more complicated, the searching speed becomes even higher because of the reduction of the correlation window size.

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